


# Mnemorph

An agent harness where trust is earned, proven and written down

# An agent can do anything, and nothing proves it did

### Power comes from a prompt

What an agent may do lives in text it can read, and often rewrite.

### Done is its own word

It says the work is finished. Nothing outside it has checked.

### Mistakes leave no trail

Logs live where the agent writes, so they can be changed afterwards.

# Mnemorph makes the rules code, not configuration

Every agent is a creature of a race. What each race may do, whom it answers to and how it rises or falls is compiled in.

### Rust core

The law is compiled. No prompt or config file can widen it.

### Kernel sandbox

Every command runs under a guard, in the soil its creature owns.

### Trusted journals

Hash-chained, kept outside the world, out of every body's reach.

# A COLONY, WITH ONE JOB FOR EACH CREATURE



## Rimuru

Understands you, plans and answers.



## Elf

Plans one lane of work.



## Orc

Writes the test first, judges the landing.



## Slime

The only race that writes code.



## Kijin

The only hands that reach outside.



## Dark Elf

Attacks the code. Proves every claim.

You stand over all of them as Veldora. Five ascended forms are earned, and can be lost.

# Nothing lands because an agent says so

**Plan**
exactly what would happen

→

**Gate**
you say yes, bound to that plan

→

**Act**
in the sandbox

→

**Probe**
read back from outside

→

**Receipt**
journaled, undoable

## The end gate

Runs the verify lines you pinned. A body cannot edit them.

## A cut run

Is probed, never replayed. If it landed, it isn't done twice.

## A changed world

Runs nothing. Your yes was for what you saw.

# The colony evolves, on trial

**Due**
stressed, desk full

→

**Draft**
a model writes traits only

→

**Prove**
on a copy of the world

→

**Trial**
10 runs vs doing nothing

→

**Keep or roll back**
by the rule

## Kept only if it wins

At least 2 more tasks won than lost, with no more problems than doing nothing.

## Never self-rewriting

The harness writes the file. Race, ground and verify lines are out of reach.

# Measured on real memory, recall needed fixing

**0.03**

### Before: F1 at 101 notes

A fixed setting let nearly every note match once memory grew.

**0.66**

### Best fit: F1 at 101 notes

The right setting grows with the square root of memory size.

### How we know

Notes written by real bench runs, each task's ask as the query. One model, small tasks: a first calibration.

# The board shows the colony, live




**The net:** a 3D city of races, bodies and Minds

**Approvals** with the terminal's protections

**Memory, ontology, metrics** for every world

**Loopback only:** nothing leaves your machine

# Honest numbers: the chain doesn't pay yet

## 6/10

### Solo

Real sqlglot fixes across modules, hidden tests passed.

## 3/10

### Hierarchy

The full chain, on the same ten tasks.

## 2.8×

### The calls

More work for fewer fixes. Most runs stopped at a checkpoint.

One run per task, one cheap model: a signal, not a proof. Small tasks, 10 paired so far: all passed, at 17× the tokens.

# Proven against the real world, not only tests

### A live forge run

A branch and a pull request through Kijin, one approval. It found a login bug the stub forge had hidden.

### CI on two systems

macOS caught a journal path bug that Linux could not show.

### A real audit

Lockfiles checked offline against OSV, in the sandbox, network off.

# Where it stands

## M0–M13

### Built

Core, sandbox, memory, the world, Kijin, Dark Elf, evolution, the board.

## 1.0

### Next

A signed release: a fresh machine running in under two minutes. The hierarchy still has to earn its cost.

The SDK and protocol are on npm: @mnemorph/sdk and @mnemorph/protocol